

Lecture 1: Introduction

COMP3018/COMP5018: Human-Robot Interaction

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Introduction

- Today's Topics

- 1- The concept of human-robot interaction
- 2- Social robotics
- 3- Cognitive robotics

- Session Learning Outcomes

By the end of today's lecture you will be able to:

- 1- Describe the different areas in the field of human-robot interaction.
- 2- Describe social robotics and its applications.
- 3- Discuss the difference between social robotics and cognitive robotics.

Introduction to COMP3018/COMP5018

- Module Aims

- Equip students with an understanding of multimodal human-robot interaction and its applications in daily life.
- Enable students to create, and evaluate, a robot system for interaction with users.
- Provide insights into the current and future challenges facing human-robot interaction.

- Learning Outcomes

- Explain the elements included in the field of human-robot interaction.
- Describe what differentiates cognitive robotics from the other sub-branches of robotics.
- Design a human-robot interaction experiment and evaluate the performance of the system through appropriate metrics and evaluation methods.

Introduction

- This module is comprised of the following indicative syllabus:

1- Human-Robot Interaction

- Introduction to Human-Robot Interaction and its Applications
- Verbal and Nonverbal Communication
- Emotional Intelligence
- Spatial Interaction
- Metrics and Evaluation Methods

2- Cognitive Robotics

- Introduction - What is Cognitive Robotics?
- Embodied Cognition
- Cognitive Architectures
- Learning from Demonstration

3- Models for Human-Robot Collaboration

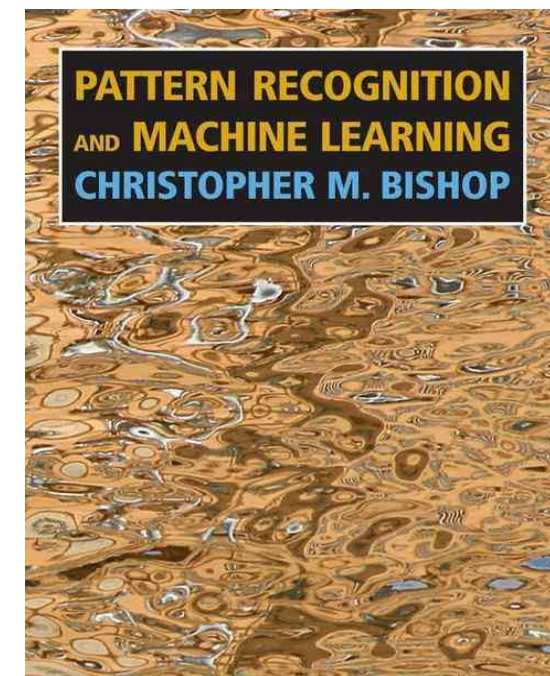
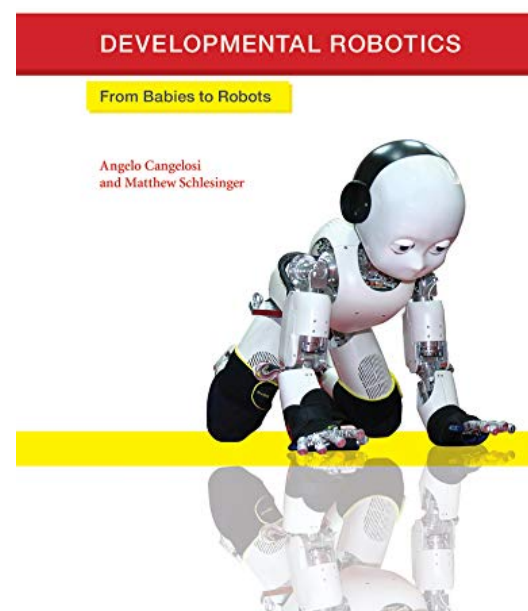
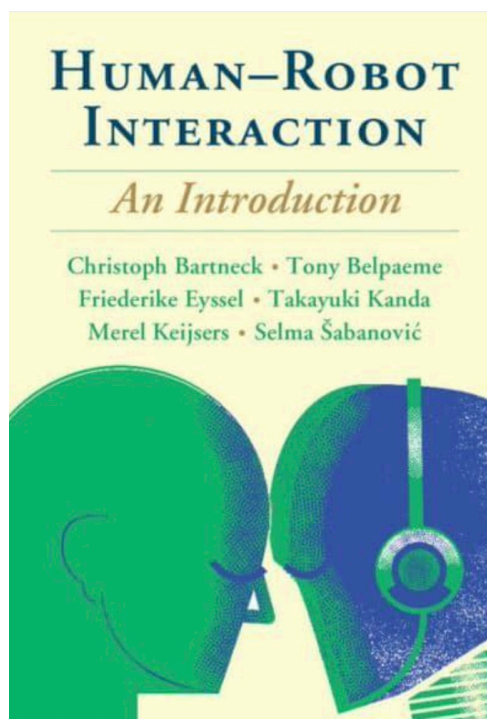
4- Ethics for Robotics and AI



Introduction

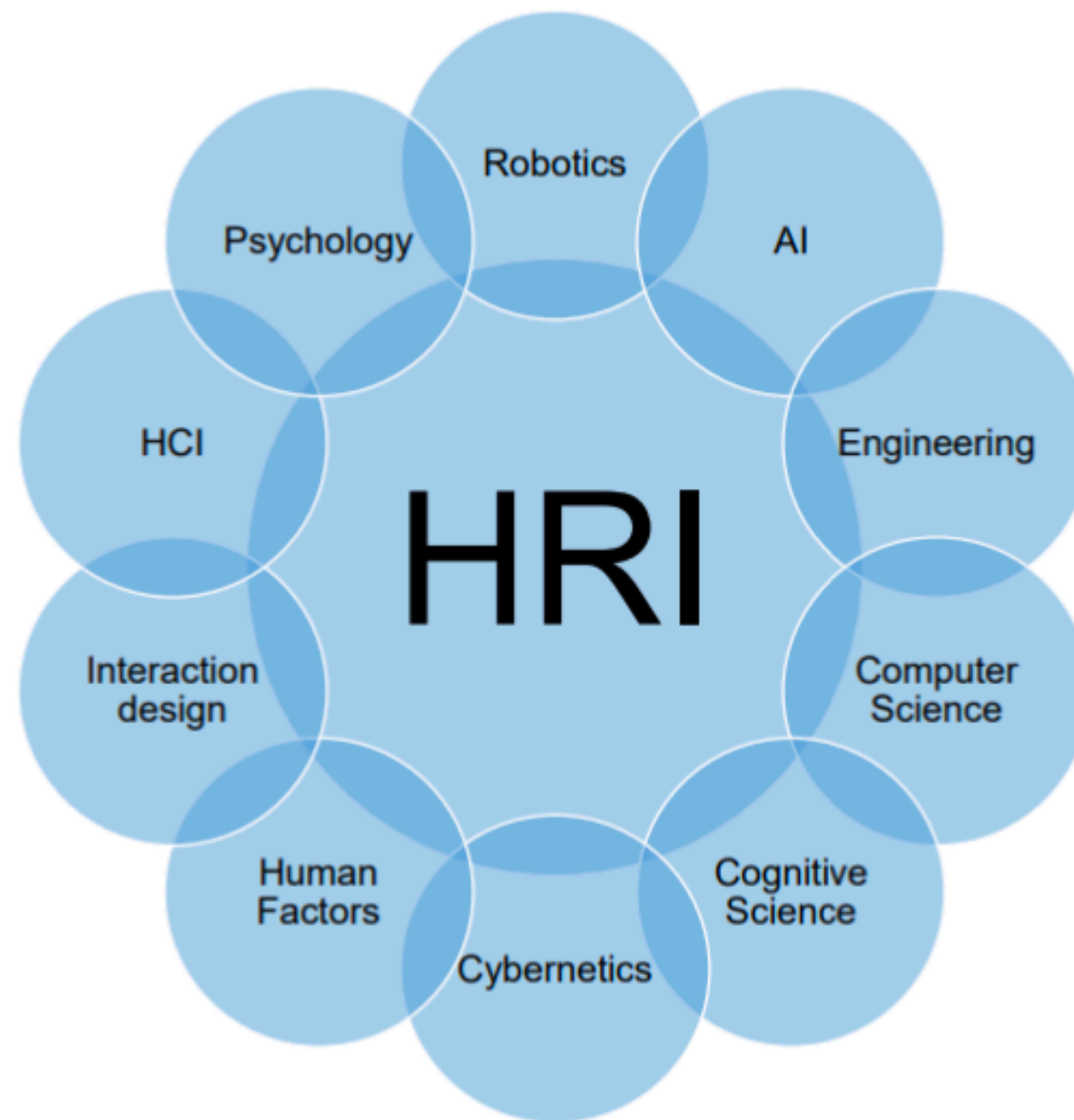
Recommended Reading Sources

1. C. Bartneck and T. Belpaeme, et al., **"Human-Robot Interaction"**, Cambridge University Press, 2020.
2. A. Cangelosi and M. Schlesinger, **"Developmental Robotics: From Babies to Robots"**, MIT Press, 2015.
3. Christopher M. Bishop, **"Pattern Recognition and Machine Learning"**, Springer, 2006.



Human-Robot Interaction

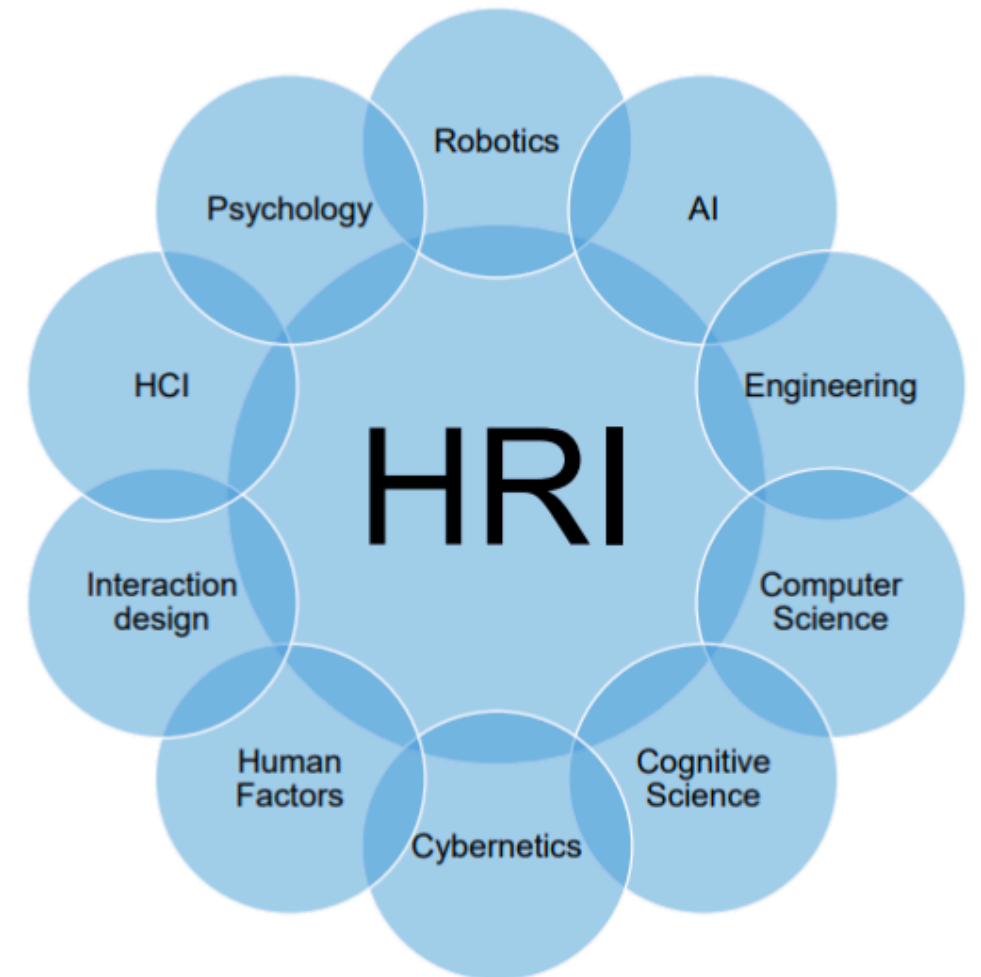
- **Human-Robot Interaction (HRI)** is a multidisciplinary field that studies interaction and collaboration between human users and robots, and how robot systems can be **designed, developed, used, and evaluated**.



Human-Robot Interaction

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“A robot must obey any order given to it by human beings”, Asimov 2nd law in 1941.



Elder Care

<https://www.asianz.org.nz/>



Store

<https://customerthink.com/>



Service

<https://global.toyota/jp/>

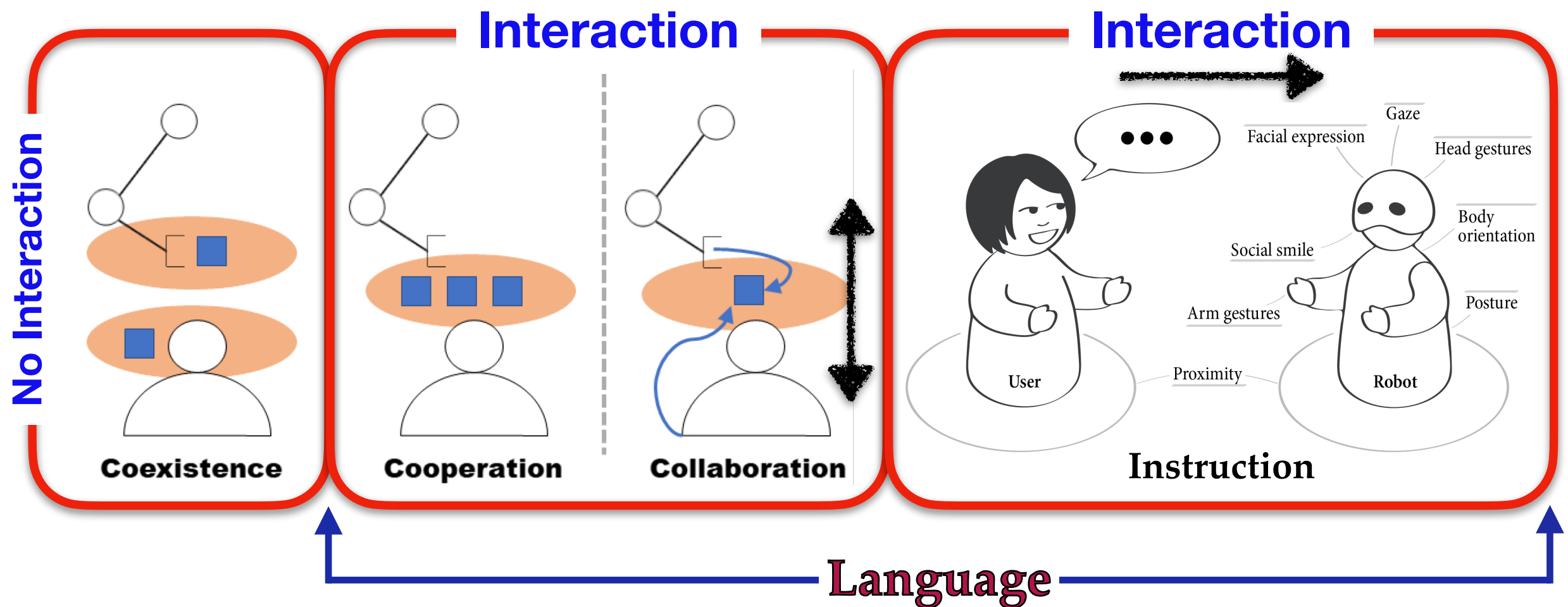


Industry

<https://www.kuka.com/>

Human-Robot Interaction

- **Human-Robot Interaction** can be characterized in terms of interdependency by:
 - ◉ **Coexistence**: same environment & different tasks (no “or minimal” interaction)
 - ✦ **Cooperation**: same workspace & complementary parts of a task (shared goal)
 - ✦ **Collaboration**: same workspace & same task (shared goal)
 - ✦ **Instruction**: same environment (workspace) & master-slave setting

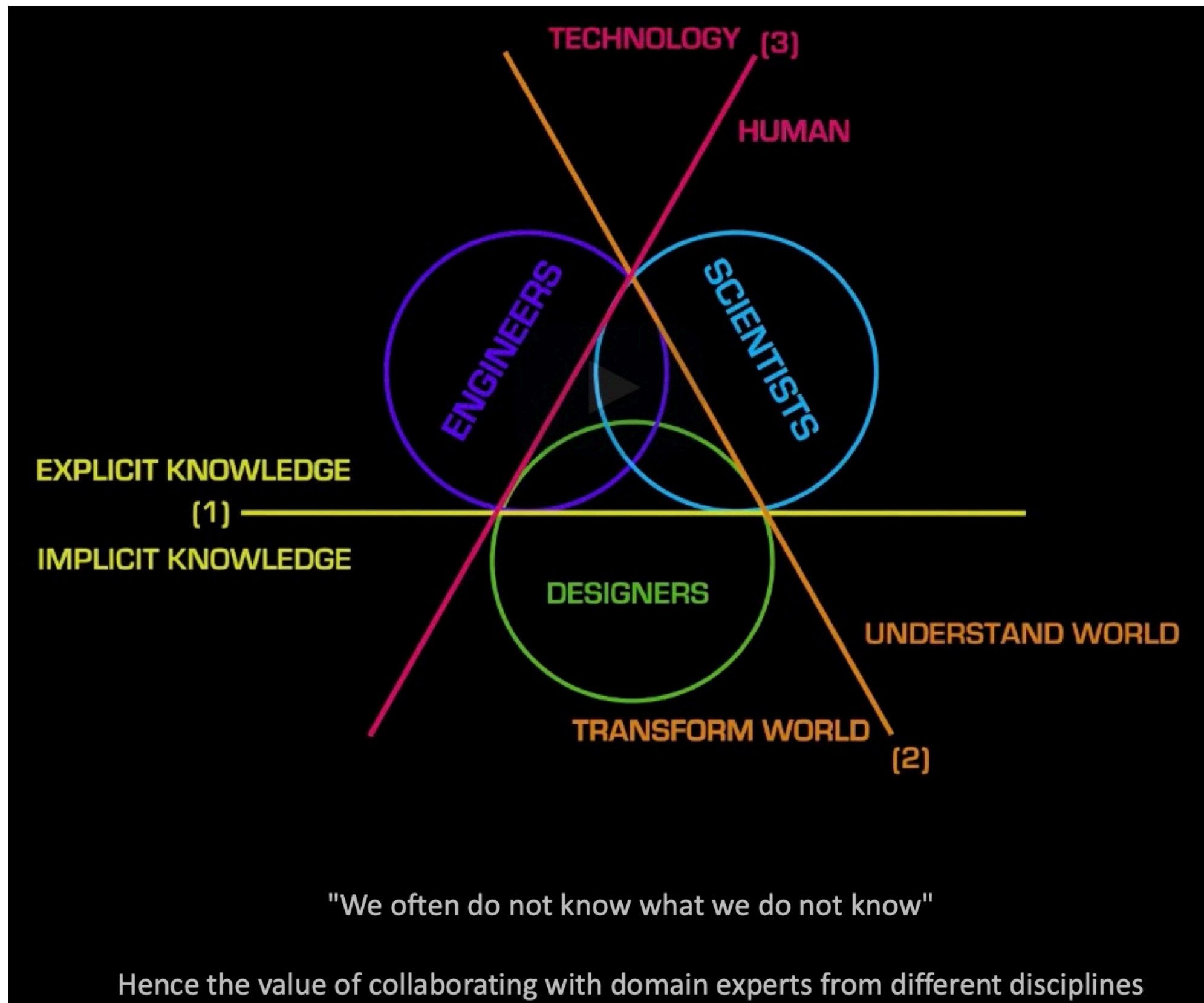


[1] E. Matheson, R. Minto, E. G. G. Zampieri, M. Faccio and G. Rosati, “Human-Robot Collaboration in Manufacturing Applications: A Review”, *Robotics*, 8(4), 2019

[2] J. Butepage and D. Kragic, “Human-Robot Collaboration: From Psychology to Social Robotics”, arXiv:1705.10146, 2017

[3] B. Mutlu, “Designing Embodied Cues for Dialog with Robots”, *AI Magazine*, 32(4):17–30, 2011

Human-Robot Interaction



What is a Social Robot?

- A social robot is an autonomous robot that **interacts and communicates** with humans or other autonomous physical agents by following social behaviors and rules attached to its role.

ROBOT

Detection layer

DETECTION



EMOTIONS

Action layer

HUMAN

What is a Social Robot?

Robots that evoke humanlike character traits:

- Exploit multiple, fast reactive behaviors to give a sense of social presence
- Give at least an appearance of intelligence



AIBO Robot
Entertainment



Paro Robot
Healthcare



Keepon Robot
Research - Education

What is a Social Robot?

Many other related definitions ...

- Assistive robotics (non-social)
- Socially assistive robotics (companion, service)
- Socially and emotionally assistive robotics
- Social robotics
- Interactive robotics
- Socially interactive robotics
- Affective robotics
- Human-related robotics
- Cloud robotics
- Telepresence robots
- Virtual agents

Where social (& social assistive) robots have been used?

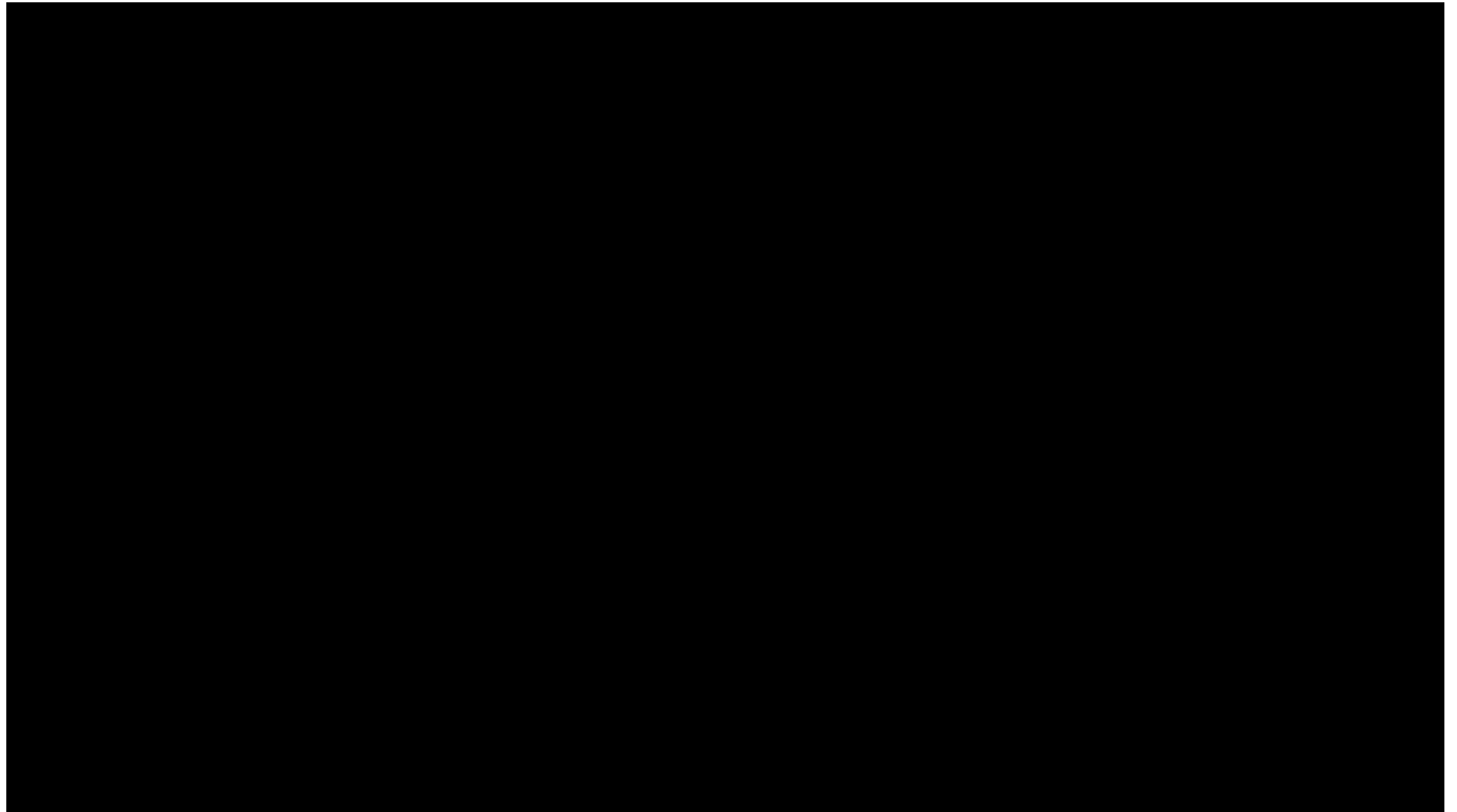
- **Elderly care**
 - assistive living, memory games
- **Physical recovery, rehabilitation, and training**
 - stroke, injuries
 - education
- **Care of individuals with cognitive and social disabilities**
 - dementia, autism
- **Entertainment**
- **Working life**
- Creating Long-term relationships: Acceptance, trust

Examples of Social Robots



Paro Robot

Examples of Social Robots



AIBO Robot

Examples of Social Robots



Pepper Robot

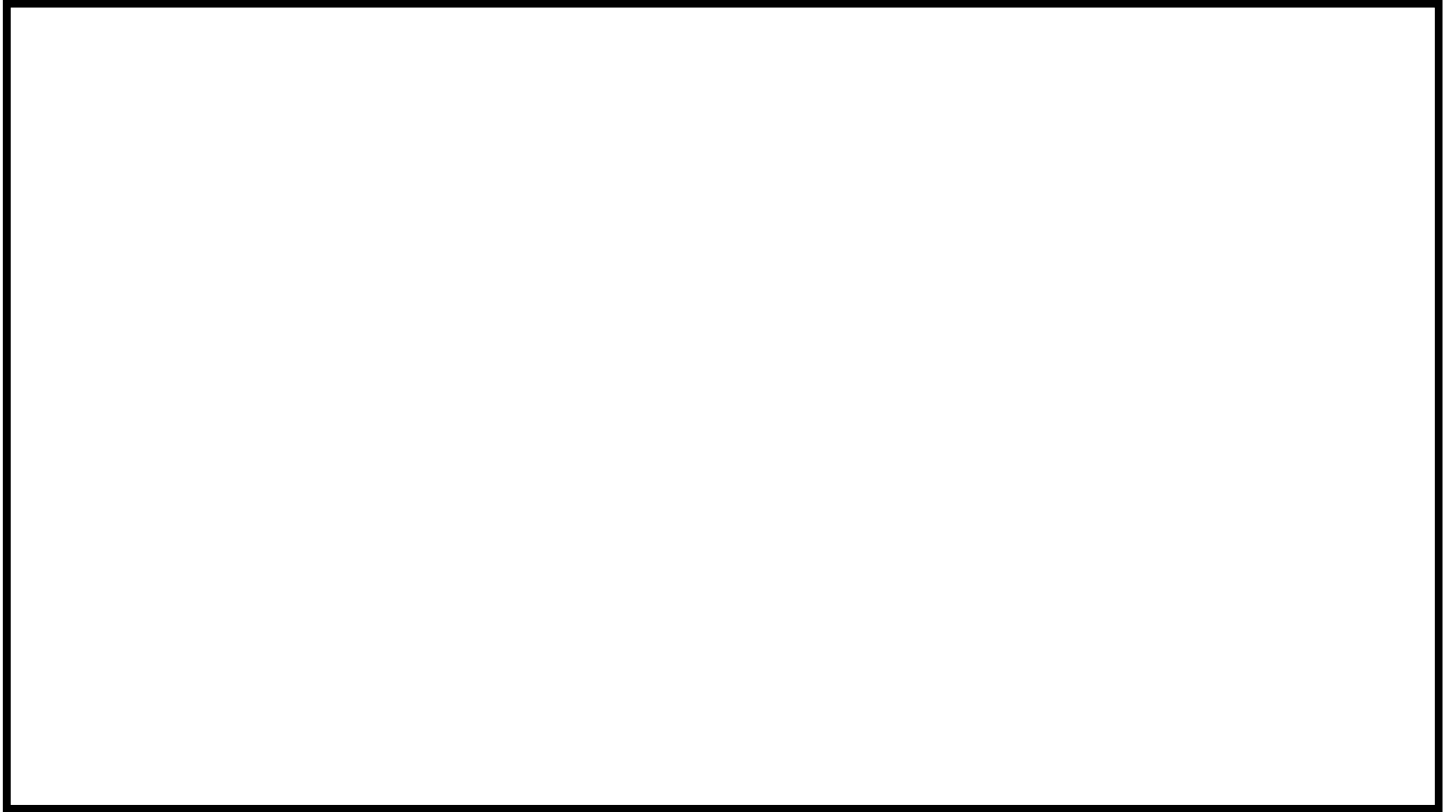
Examples of Social Robots



Furhat Robotics

Furhat Robot

Examples of Social Robots???



HSR Robot

Examples of Social Robots???



Geminoid Robot

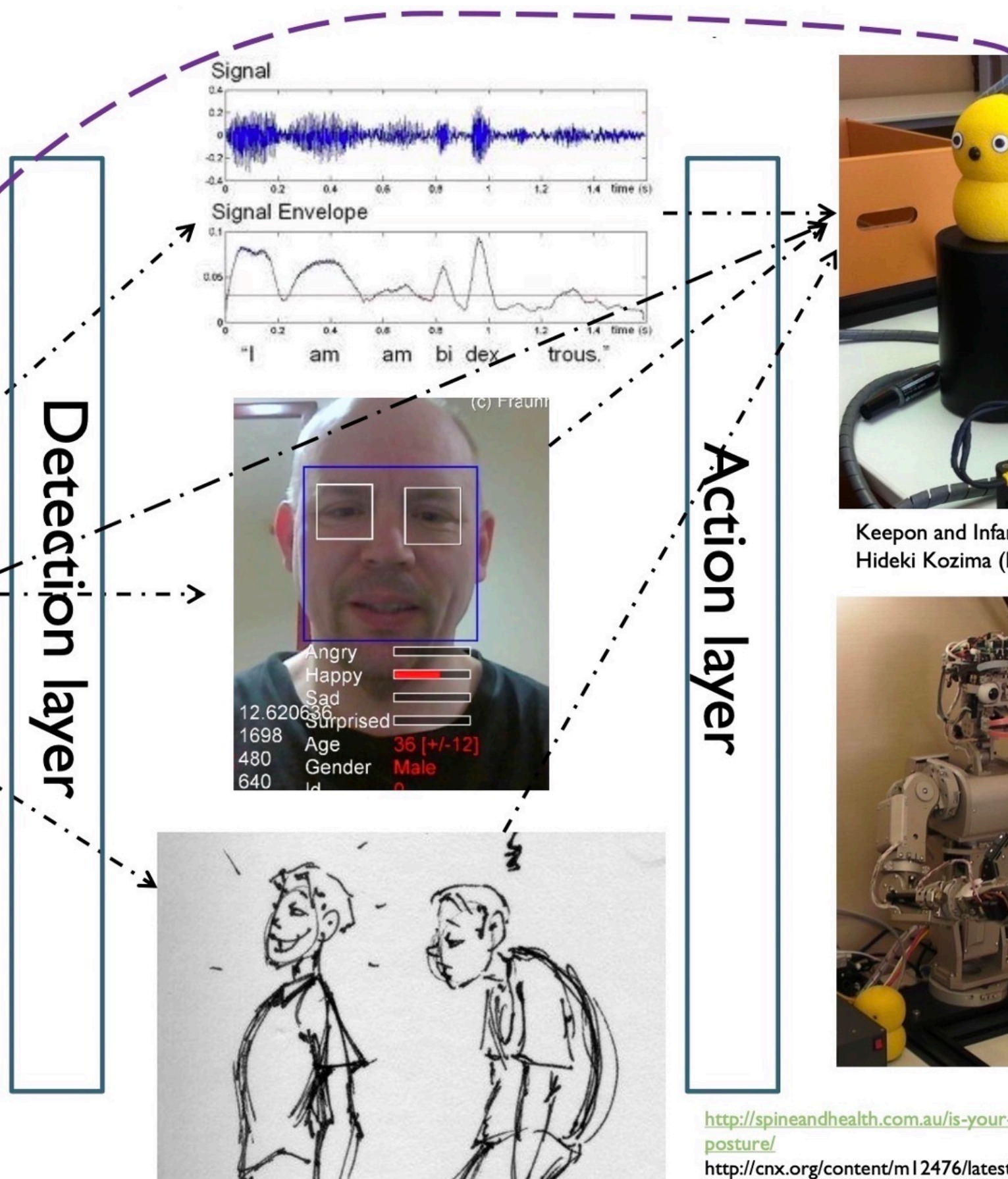
Loop from Human to Robot (HRI)

Where stress/depression or emotions are located?
Is it possible to quantify those severeness?



Are we able to construct a loop between emotions and robot actions?

How to measure or quantify possible effects?



Some Possibilities and Challenges

- To use the robot as a motivator and a coach
- To use the robot as an interactive channel for detecting early symptoms
- To use the robot as an observer without human-related subjective bias
- To use the robot as an information source and co-worker
- To use the robot as a safety & security partner

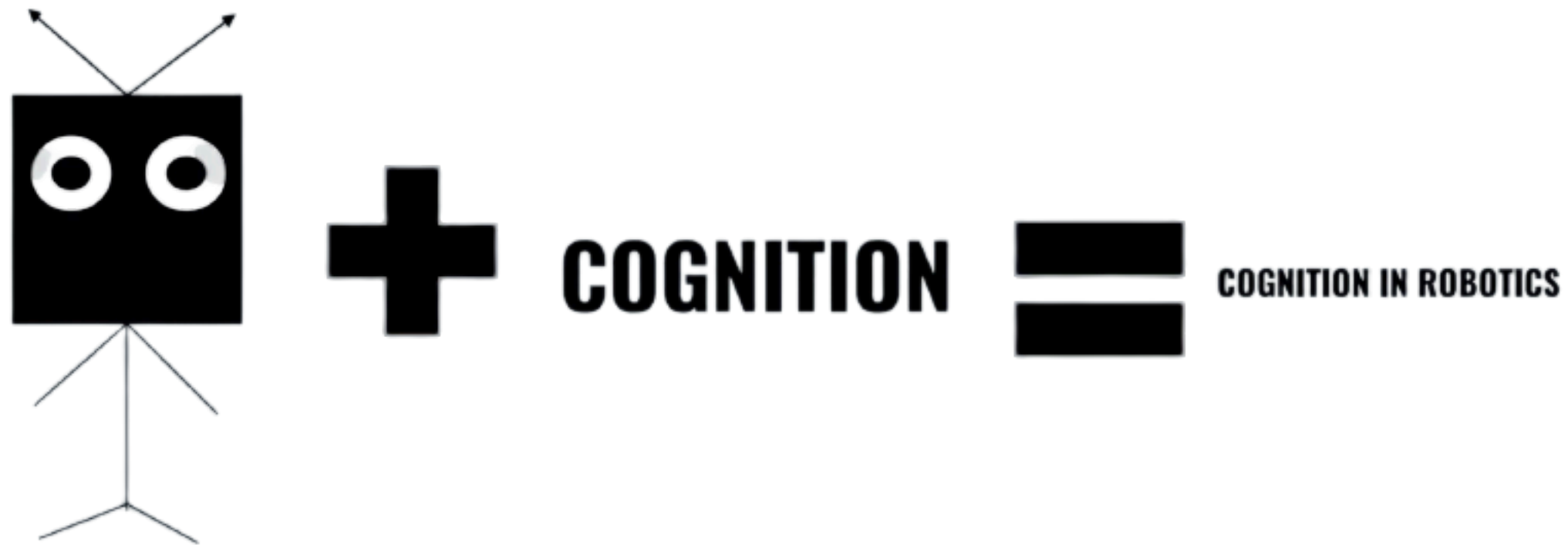
However ...

- How do people perceive robots?
- Use a virtual agent or physical robot or mix?
- Having an industrial robot that is interactive and social?
- What kind of services will be available?

Cognitive Robotics

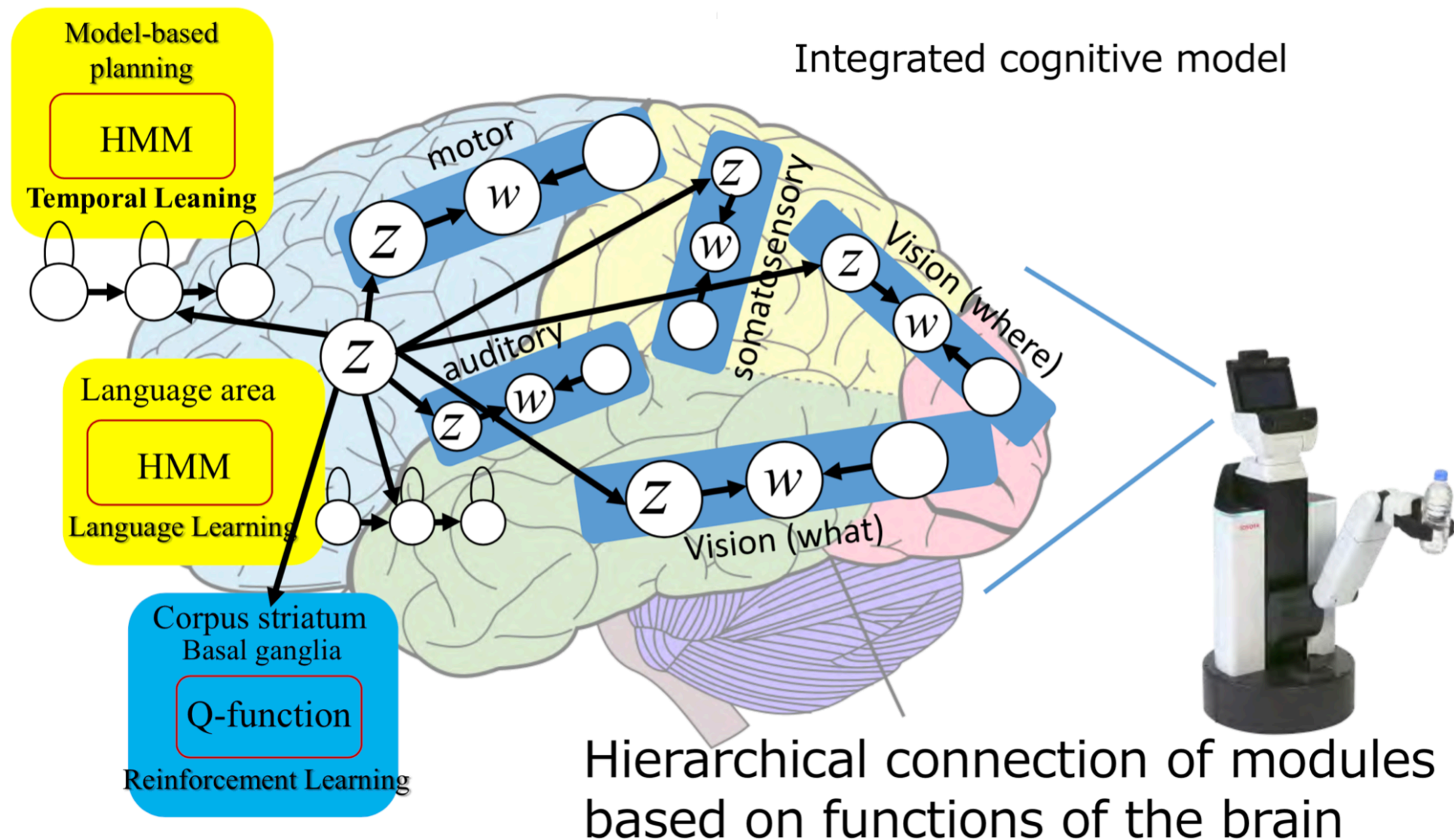
- What is cognition

It is the process of acquiring knowledge and understanding through thoughts, experiences, and senses.



Cognitive Robotics

- We need an integrated cognitive model that can learn from multimodal information.



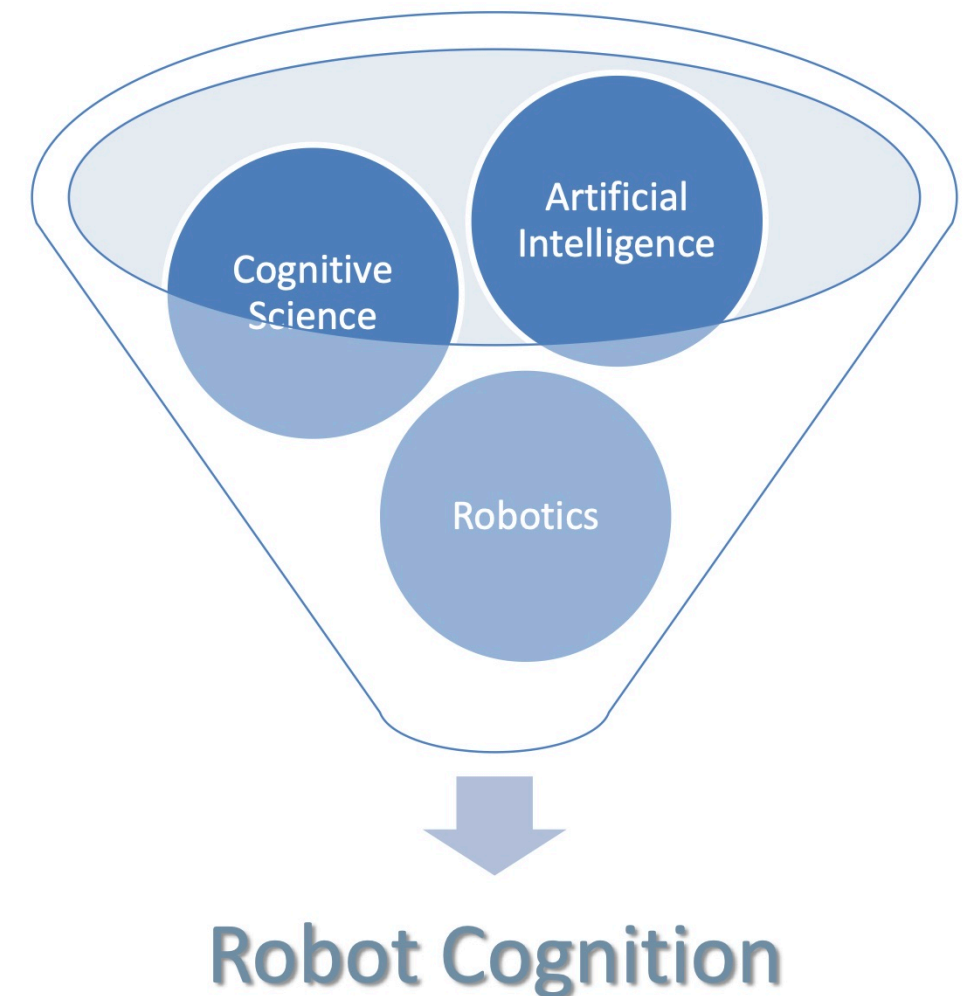
- The striatum coordinates multiple aspects of cognition, including action planning, **decision-making**, and reward perception.

Cognitive Robotics

Improving robots capabilities with cognition

Cognitive Robotics provides cognitive building blocks to robots:

- High-Level Perception and Action
- Attention
- Memory
- Learning
- Concept Formation
- Reasoning and Problem Solving
- Communication and Use of Language
- Theory of Mind
- Social Interaction
- ...



Cognitive Science to Robotics

- Cognitive Robotics as applying knowledge from cognitive science to robotics.
 - Cognitive psychology has described cognitive capabilities of humans
 - Cognitive science tries to build computational models that could be implemented in a robot
- But why should robots work the way humans work?
 - Maybe robots can do things better than humans!
- On the other hand, maybe there are good reasons for why humans work the way they do, and we can learn from that for robotics
 - Cognitive Science tries to understand why human cognitive capabilities are as they are

Robotics to Cognitive Science

- Cognitive Robotics can be used to drive the science of cognition
 - Cognitive robotics as a platform to test theories about human cognition.
 - Cognitive Science as the science of all forms and kinds of cognition and cognitive agents, whether these be human, animal, alien, robot, or otherwise. Thus:
 - Robotic systems are cognitive systems, and are interesting to study in and of themselves

Robotics to Cognitive Science

- Cognitive Robotics as Experimental Cognitive Science
 - Cognitive Robotics as the use of robotics to explore cognitive systems or architectures, to develop new concepts and frameworks of cognition, and to formulate and test cognitive hypothesis.

Cognitive Robotics



Affordances learning

Summary

- Today we discussed

- 1- The concept of human-robot interaction
- 2- Social robotics
- 3- Cognitive robotics

- Next week

- 1- We will continue discussing a different topic in social robotics; i.e., verbal and non-verbal communication.